

## Recursive Method:

A recursive method is a method in which a function calls itself to solve smaller versions of the same problem until a base condition is reached.

## Algorithm:

Step 1: Start

Step 2: Read n from the user

Step 3: Define function factorial(n)

Step 4: If  $n == 0$  or  $n == 1$ , return 1

Step 5: Otherwise, return  $n \times \text{factorial}(n-1)$

Step 6: Display the result

Step 7: Stop

## Out put:

```
Enter a number: 3
Factorial = 6
Execution Time = 1.2341999990894692e-05 seconds
Time Complexity = O(n)

...Program finished with exit code 0
Press ENTER to exit console.
```

**Time Complexity:  $O(n)$**

**Space Complexity:  $O(n)$**